NEW YORK CITY AIRBNB MARKET ANALYSIS

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1. Introduction

Airbnb marketplaces generate large amounts of listing data. Analysing this data helps understand where listings are located, which accommodation types are offered, how prices differ, how hosts manage their listings and how guests respond through reviews. This project uses Microsoft Power BI to transform New York City Airbnb listing data into an interactive business intelligence dashboard.

The project follows a complete analysis process: data preparation in Power Query, feature engineering, calculation using DAX, dashboard creation, interpretation of findings and recommendations.

The final dashboard includes Market Overview, Location Analysis, Room & Pricing, Host Performance, and Reviews & Availability pages. Interactive Borough, Room Type and Price Category slicers and navigation buttons were also added to make the report easier to explore.

2. Project Objective

- To understand how listings are distributed across boroughs and neighbourhoods.
- To compare room types by price and minimum-stay requirement.
- To understand the host portfolio structure and host performance.
- To study how reviews vary by location and over time.
- To study how available listings are throughout the year.
- To create price, availability, review-time and host-portfolio categories.
- To present the analysis through an interactive Power BI dashboard.

In addition, the project aims to ensure data quality by identifying and removing duplicate listing records, correcting invalid values, and using interactive filters to analyse the dashboard from different perspectives.

3. Dataset Description

The project uses the Kaggle New York City Airbnb Open Data (AB_NYC_2019.csv). The working query used in the report is referred to as 'copy_airbnb'.

Field	Type	Purpose
id	Whole number	Identifies a listing
name	Text	Listing name
host_id	Whole number	Identifies a host
host_name	Text	Host name
neighbourhood_group	Text	Borough
neighbourhood	Text	Neighbourhood
latitude	Decimal	Used for the map
longitude	Decimal	Used for the map
room_type	Text	Type of accommodation
price	Whole number	Price used in price calculations
minimum_nights	Whole number	Minimum stay required
number_of_reviews	Whole number	Number of reviews
last_review	Date	Date of the latest review
reviews_per_month	Decimal	Monthly review activity
calculated_host_listings_count	Whole number	Number of listings of the host
availability_365	Whole number	Days available in a year

The working source contained 49,080 rows and 16 columns. During the data preparation stage, the dataset was checked for duplicate records, missing values, incorrect data types, invalid values and extreme values. After cleaning, the report shows 48,715 listings.

Some listings have very high prices or very high minimum-night values. These records were reviewed and retained, because they may represent genuine premium or long-stay listings.

4. Data Preparation and Power Query

Power Query was used to prepare the raw listing data before building the dashboard.

- Promoted the first row as headers.
- Assigned suitable data types to the columns.
- Trimmed text values for consistency.
- Checked column quality and errors.
- Removed 13 duplicate rows after checking the id column (49,080 rows contained 49,067 distinct listing IDs).
- Replaced null values in reviews_per_month with 0, because no monthly review activity was recorded for those listings.
- Replaced null values in name and host_name with Unknown to avoid blank labels.
- Removed rows where price was 0, because they are not meaningful for price analysis.
- Removed rows where price contained errors (text such as the room type had moved into the price column).

- Removed error rows from id, host_id, longitude, number_of_reviews and last_review.
- Retained null values in last_review (about 21% of rows), because a missing date means no review was recorded. Replacing them with an artificial date would distort the time analysis.
- Reviewed high prices (up to 10,000) and high minimum-night values and retained them as possible premium and long-stay listings.

Additional columns were created to make the analysis easier to read:

Derived Field	Rule	Purpose
Price Category	Up to 300 Budget; up to 1000 Standard; up to 5000 Premium; above 5000 Luxury	Price segmentation
Availability Category	Up to 90 Low; up to 180 Medium; up to 300 High; above 300 Very High	Availability segmentation
Review Year	Year from last_review	Trend analysis
Review Month	Month from last_review	Monthly analysis
Review Quarter	Quarter from last_review	Quarterly analysis
Host Portfolio Size	1; 2-5; 6-10; 11-20; 21+ listings	Host scale
Host Portfolio Sort	Numeric key 1 to 5	Correct display order

5. Data Model and Calculations

The cleaned 'copy_airbnb' table is the main source for the report. DAX measures are calculated from this table, allowing the Borough, Room Type and Price Category slicers to update the visuals.

Main measures include Total Listings, Total Hosts, Average Price, Maximum Price, Total Reviews, Average Reviews per Listing, Average Availability, Average Minimum Nights and Average Listings per Host.

The report was designed so that the three slicers interact with the dashboard visuals. When a slicer selection is changed, the related KPIs and charts update according to the selected filter context.

The DAX measures were created in the Power BI model/report layer. They are not physical columns in Power Query. Power Query was used for data preparation, while DAX was used for calculations and interactive analysis after the data was loaded into the model.

6. DAX Functions and Measures

Function	Purpose	Project use
DISTINCTCOUNT	Counts unique values	Listings and Hosts
SUM	Adds column values	Total Reviews
AVERAGE	Calculates an average	Price, Availability, Minimum Nights
MAX	Returns the largest value	Maximum Price
DIVIDE	Safe division	Reviews per Listing, Listings per Host

Important DAX Measures

Measure	DAX Formula
Total Listings	`DISTINCTCOUNT(copy_airbnb[id])`
Total Hosts	`DISTINCTCOUNT(copy_airbnb[host_id])`
Average Price	`AVERAGE(copy_airbnb[price])`
Maximum Price	`MAX(copy_airbnb[price])`
Total Reviews	`SUM(copy_airbnb[number_of_reviews])`
Average Reviews per Listing	`DIVIDE([Total Reviews], [Total Listings])`
Average Availability	`AVERAGE(copy_airbnb[availability_365])`
Average Minimum Nights	`AVERAGE(copy_airbnb[minimum_nights])`
Average Listings per Host	`DIVIDE([Total Listings], [Total Hosts])`

Slicers: Borough, Room Type and Price Category

KPI	Value	Meaning
Total Listings	About 49K	Cleaned listing volume
Total Hosts	About 37K	Distinct hosts
Average Price	$152.74	Average listing price
Total Reviews	About 1M	Total review activity
Average Reviews per Listing	23.25	Engagement benchmark
Average Availability	112.61 days	Availability benchmark
Average Minimum Nights	7.04	Stay requirement benchmark
Average Listings per Host	1.30	Host portfolio benchmark

7. Market Overview Analysis

- Total Listings: about 49K
- Total Hosts: about 37K
- Average Price: $152.74
- Average Availability: 112.61 days
- Total Reviews: about 1M
- Manhattan and Brooklyn have the largest number of listings.
- Entire homes/apartments and private rooms account for most listings.
- Budget listings are by far the largest price category.

This page gives the overall market picture by combining KPI cards with borough, price category, average price and room type visuals. The Borough, Room Type and Price Category slicers filter every visual on the page.

8. Location Analysis

- Total Listings: about 49K
- Average Reviews per Listing: 23.25
- Average Price: $152.74

- Williamsburg and Bedford-Stuyvesant have the most listings among neighbourhoods.
- Fort Wadsworth and Woodrow have the highest average prices in the displayed Top 10.
- Listings are concentrated in the central areas of New York City on the map.

This page explains the geography of the market. It combines a map of listing locations with neighbourhood price and listing-count rankings and the room type mix of each borough.

9. Room & Pricing Analysis

- Average Minimum Nights: 7.04
- Average Price: $152.74
- Maximum Price: 10K
- Entire homes/apartments have the highest average price (about $210).
- Private rooms average about $90 and shared rooms about $70.
- Entire homes/apartments have the highest average minimum-night requirement.

This page compares accommodation types by price and stay requirement. The Price vs Minimum Nights scatter chart shows that most listings are grouped at lower prices and shorter stays, while a few listings sit far from the main group.

10. Host Performance Analysis

- Total Hosts: about 37K
- Average Listings per Host: 1.30
- Average Reviews per Listing: 23.25
- Most hosts have only one listing; 2-5 listings is the next largest group.
- Sonder (NYC) has the highest listing count among the displayed hosts.
- Dona has the highest review count among the displayed hosts.

This page shows how hosts are structured and how guests respond to them. It uses listing count, review count and portfolio size together, because one measure alone does not describe host performance.

11. Reviews & Availability Analysis

- Total Reviews: about 1M
- Average Reviews per Listing: 23.25
- Average Availability: 112.61 days
- Average Minimum Nights: 7.04
- Brooklyn has the highest review total, with Manhattan close behind.
- Shared rooms have the highest average availability (about 160 days).
- The Low Availability category is the largest availability group.

This page connects guest engagement with how available listings are. The Reviews vs Availability scatter chart does not show a simple rule that more availability gives more reviews.

12. Insights and Findings

- The market contains about 49K listings managed by about 37K hosts, with an average price of $152.74.
- Listings are concentrated in Manhattan and Brooklyn.
- Budget listings dominate the price categories.
- Entire homes/apartments have the highest average price among room types.
- Most hosts have only one listing, so the market is mainly made up of individual or small-scale hosts.
- Average reviews per listing is 23.25, but reviews vary by borough and by listing.
- Average availability is 112.61 days, and shared rooms have the highest average availability.

- Some prices and minimum-night values are far above the main group and should be checked before being treated as typical.

The dashboard shows descriptive patterns. Listing counts, review totals and availability do not by themselves prove customer demand, bookings or profitability.

13. Recommendations

- Protect the strongest markets by maintaining listing quality and host support in Manhattan and Brooklyn.
- Investigate Queens, the Bronx and Staten Island using visitor demand and booking data before deciding on host-acquisition campaigns.
- Use neighbourhood-level price benchmarks together with listing count and room type, instead of relying only on the city-wide average.
- Show room type choices clearly, because entire homes cost more on average than private and shared rooms.
- Support single-listing hosts with simple guidance on listing quality, pricing and review responses.
- Validate extremely high prices and long minimum stays. Keep valid premium listings and correct any data errors.
- Add booking, occupancy and revenue data in future work to measure profitability.

14. Conclusion

This project demonstrates how Power BI can convert raw Airbnb listing data into an interactive business intelligence dashboard. Power Query was used for preparation and DAX was used for analysis. The analysis helped identify:

- Where listings are concentrated
- How prices differ by room type and neighbourhood
- How hosts are structured
- How reviews and availability vary across the market
- Areas where the analysis can be extended

The final Power BI report provides a complete view of the New York City Airbnb market. The interactive design allows users to filter the analysis by Borough, Room Type and Price Category and observe the corresponding changes in KPIs and visuals.

Appendix: Dashboard

The final Power BI dashboard consists of five interactive pages:

1 Market Overview

2 Location Analysis

3 Room & Pricing

4 Host Performance

5 Reviews & Availability

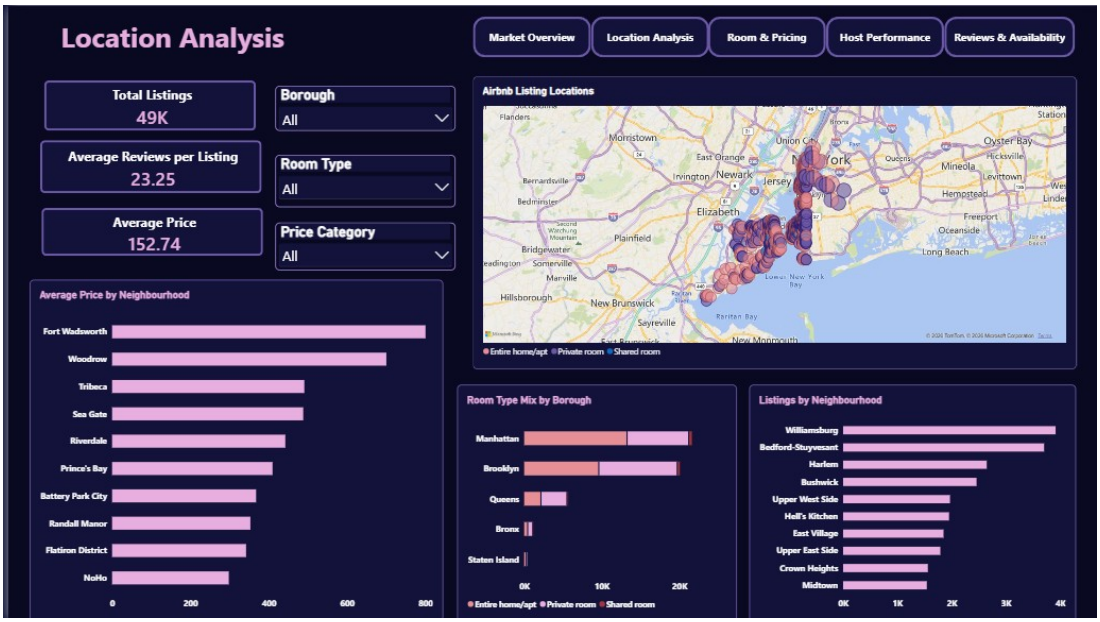
The dashboard includes interactive Borough, Room Type and Price Category slicers, KPI cards, charts and navigation buttons. These features allow the user to explore the dataset from different perspectives and understand the major business findings. The insights from each page are given below its screenshot.

Market Overview – Page 1



- KPI cards: The dashboard shows about 49K listings, 37K hosts, an average price of $152.74, average availability of 112.61 days and about 1M total reviews.
- Listings by Borough: Manhattan has the most listings (about 21.7K), followed by Brooklyn (about 20K). Queens has far fewer, and the Bronx and Staten Island have the smallest counts. The chart shows concentration but cannot prove the reasons for it.
- Listings by Price Category: Budget listings are the largest group by a wide margin. Standard, Premium and Luxury listings are much smaller groups.
- Average Price by Borough: Manhattan has the highest average price, followed by Brooklyn and Staten Island. Queens and the Bronx are lower.
- Room Type Distribution: Entire homes/apartments and private rooms make up most listings. Shared rooms are a very small share.

Location Analysis – Page 2



- KPI cards: Total Listings is about 49K, Average Reviews per Listing is 23.25 and Average Price is $152.74.
- Airbnb Listing Locations: Listings cluster strongly around the central areas of New York City. The map shows where listings are located, not bookings or demand.
- Average Price by Neighbourhood: Fort Wadsworth and Woodrow are at the top of the displayed Top 10, followed by Tribeca and Sea Gate. Neighbourhoods with few listings can give less stable averages.
- Room Type Mix by Borough: Manhattan and Brooklyn have the largest volumes, mostly entire homes/apartments and private rooms. The Bronx and Staten Island have very few listings.
- Listings by Neighbourhood: Williamsburg and Bedford-Stuyvesant lead, followed by Harlem and Bushwick.

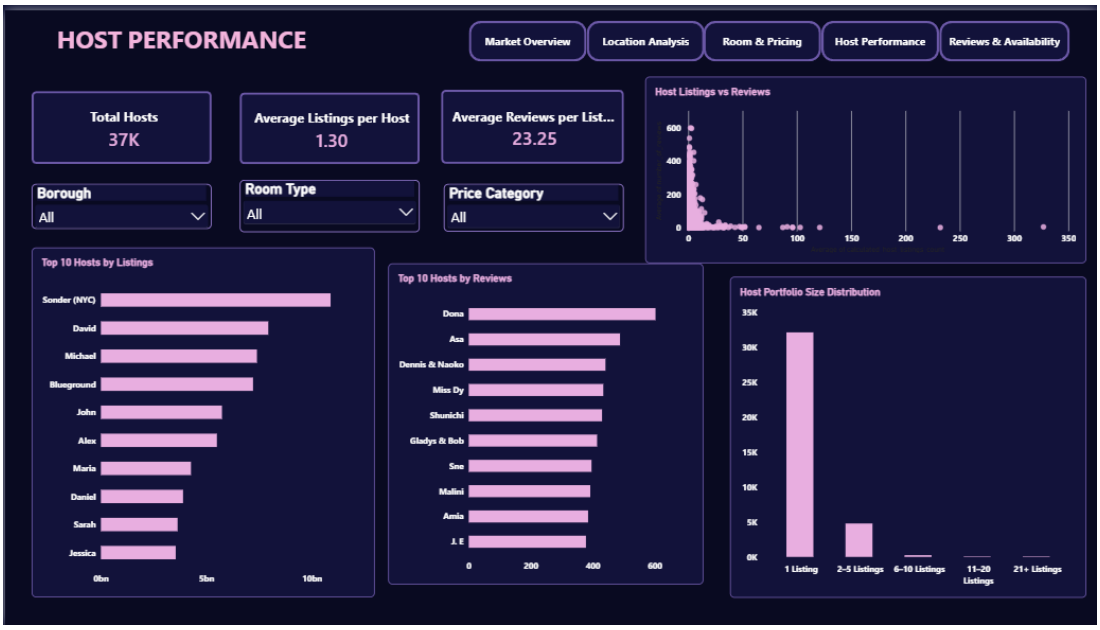
Room & Pricing – Page 3



- KPI cards: Average Minimum Nights is 7.04, Average Price is $152.74 and Maximum Price is 10K. The maximum is an extreme value and not a typical nightly price.
- Price vs Minimum Nights: Most points are grouped at lower prices and shorter minimum stays. A few points are far from the main group and are worth checking.
- Average Minimum Nights by Room Type: Entire homes/apartments have the highest average, followed by shared rooms and then private rooms.

- Room Type by Price Category: Entire homes/apartments and private rooms make up most listings in every category, and Budget is the largest category. This chart compares listing counts, not average prices.
- Average Price by Room Type: Entire homes/apartments have the highest average price (about $210), private rooms are lower (about $90) and shared rooms are the lowest (about $70).

Host Performance – Page 4



- KPI cards: The page shows about 37K hosts, 1.30 average listings per host and 23.25 average reviews per listing.
- Top 10 Hosts by Listings: Sonder (NYC) has the highest bar, followed by David and Michael. The same name can belong to more than one host, so host_id is the safer identifier.
- Top 10 Hosts by Reviews: Dona is highest, followed by Asa and Dennis & Naoko. A host can have many reviews without having the largest portfolio.
- Host Portfolio Size Distribution: Most hosts are in the 1 Listing group, and 2-5 Listings is the next largest. Larger portfolios are very small groups.
- Host Listings vs Reviews: Most points are at low listing and review counts, with a few high-count outliers. Use the tooltip to see which host each point represents.

Reviews & Availability – Page 5



- KPI cards: The page shows about 1M total reviews, 23.25 average reviews per listing, 112.61 average availability days and 7.04 average minimum nights.
- Reviews by Borough: Brooklyn has the highest review total, with Manhattan close behind. Queens is much lower, and the Bronx and Staten Island are the lowest. Review totals are affected by the number of listings.
- Average Availability by Room Type: Shared rooms have the highest average availability (about 160 days), while entire homes/apartments and private rooms are about 110 days.
- Listings by Availability Category: Low Availability is the largest group, followed by Very High, High and Medium. These are custom categories (Low up to 90 days, Medium up to 180, High up to 300 and Very High above 300).
- Reviews vs Availability: Most listings have lower review counts across a wide range of availability. The chart does not show that more availability always gives more reviews.
- Review Activity Over Time: The line rises sharply near 2019. It uses the last_review year and the sum of number_of_reviews, so it shows review activity grouped by last-review year, not every review written each year.